Midterm · Plan the day

Week 4 · Session 3

90 minutes · individual · understanding, not memory

What the midterm checks

- Can you **read** code and explain *why* it behaves as it does?
- Can you **write** a rule and a records function that keep their promises?
- Can you **test** code so that bugs are caught?
- Can you **judge** AI-written code and name who it harms?
- Can you **explain** a decision?

The situation

- Volunteers carry food in bicycle baskets: about **15 portions** each.
- Before they leave, the coordinator needs: approved portions, **volunteers needed**, offers still **waiting**.

```
{"id": "F003", "food": "Bananas", "portions": 40, "approved": False}
```

Six parts · 100 points

Part	Task	Points
A	Read five snippets and explain *why*	20
B	`volunteers_needed(portions)` with a constant	20
C	`plan_day(offers)` returns a summary	25
D	Tests that catch planted bugs	15
E	Judge AI-written code	10
F	Your decision + clean run	10

Part D: tests that catch bugs

```
fr.catch_bugs([test_normal, test_boundary,
               test_waiting, test_empty])
```

- Runs your tests on **your** code, then on 4 versions with a **planted bug**.
- A good test **fails when the code is wrong**.
- Then it changes the decision: tests with a hidden `15` will break.

Part F: your decision

- Bicycles ~15, on foot ~8, motorbike ~30: choose for **your** campus.
- Change the constant → **Runtime** → **Restart session and run all**.
- Everything should still pass. If not, a number was hidden somewhere.
- Decision log: your choice · why (who is affected) · what happened when you reran.

Rules

- **Allowed:** your notes, the weekly notebooks, Python Basics.
- **Not allowed:** AI tools, messaging, sharing code.
- Submit the notebook as announced, after **Runtime → Restart session and run all**.
- Then a **3-minute oral check**: explain one line of `plan_day` and one test.

How to prepare

- Week 1: `//` and `%` (full loads and what is left over).
- Week 2: branch order and boundaries (`>` vs `>=`).
- Week 3: loop over records, add up, collect IDs.
- Week 4: `return` not `print`; tests for normal, boundary and empty cases.

Good luck!

Read carefully · explain *why* · test the edges · name who is harmed